

Sweet Syndrome

A constellation of symptoms and findings: the acute onset of fever, neutrophilia, erythematous, and tender skin lesions that typically show an upper dermal infiltrate of mature neutrophils.

Cardiovascular, central nervous system, gastrointestinal, hepatic, musculoskeletal, ocular, oral, otic, pulmonary, renal, and splenic organs can be involved.

Clinical Subtypes

- **Classic (idiopathic) Sweet syndrome:** May be associated with infection of the upper respiratory and/or gastrointestinal tract, inflammatory bowel disease, and pregnancy.
- **Malignancy-associated Sweet syndrome:** Occurs in individuals with previously undiagnosed or relapsing hematologic malignancies (e.g., acute myeloid leukemia) and solid tumors; presents as a cutaneous paraneoplastic syndrome.
- **Drug-induced Sweet syndrome:** Develops after initiation of certain medications—most commonly granulocyte-colony stimulating factor (G-CSF), but also all-trans retinoic acid, antibiotics, and others.

Pathogenesis

Cytokines (e.g., granulocyte colony-stimulating factor, interleukin-3, tumor necrosis factor-alpha) may play a key role in the recruitment and activation of neutrophils, contributing to the inflammatory cascade seen in the skin and systemic manifestations.

Clinical Features

Cutaneous Lesions

Skin lesions typically appear as tender, red, or purple-red papules, nodules, or plaques. The eruption may present as a single lesion or multiple lesions, often distributed asymmetrically. Lesions may coalesce into larger plaques and can be accompanied by vesicles, pustules, or bullae.

Extracutaneous involvement may affect: - Musculoskeletal (arthralgia, myalgia) - Ocular (conjunctivitis, uveitis) - Oral (ulcers) - Gastrointestinal, pulmonary, renal, and other organ systems

Diagnosis

Based on clinical, laboratory, and histopathological findings. Major criteria include: - Abrupt onset of painful erythematous plaques or nodules - Histopathology showing a dense dermal infiltrate of mature neutrophils without vasculitis

Minor criteria: - Preceding infection, pregnancy, malignancy, or drug exposure - Fever ($>38^{\circ}\text{C}$) - Response to systemic corticosteroids - Elevated erythrocyte sedimentation rate, C-reactive protein, or neutrophilia

Treatment

First-line systemic agents:

- **Corticosteroids** (e.g., prednisone 0.5–1.5 mg/kg/day): Rapid and effective response
- **Colchicine**: Alternative for mild cases or steroid-sparing
- **Potassium iodide**: Effective for some patients with localized or refractory disease

Local therapy:

- **Topical high-potency corticosteroids**
- **Intralesional corticosteroid injections:** Useful for localized or persistent lesions

Relapse may occur with tapering of therapy, particularly in malignancy-associated cases. Underlying conditions (e.g., infection, malignancy) should be investigated and managed appropriately.